

Feasibility analysis of participate in UTC calculations using Beidou PPP time link

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Abstract. In order to promote the BeiDou time transfer link to be used for UTC calculation as soon as possible, the relative calibration test of the BDS PPP link between the UTC(NTSC) and the UTC(JATC) was carried out in the National Time Service Center, and its accuracy was evaluated with reference to the GPS PPP link results published by the BIPM. The test results show that after the BDS PPP link between the UTC(NTSC) and the UTC(JATC) is calibrated, the peak deviation between the time comparison results and the BIPM GPS PPP link results is no more than 0.2 ns, satisfying the accuracy requirement of GNSS time transfer.

Keywords: UTC, BDS precise point positioning (BDS PPP), GPS precise point positioning (GPS PPP), time comparison; relative calibration

1 Introduce

The current international standard time is Coordinated Universal Time (Coordinated Universal Time, UTC). Long-distance high-precision time transfer is one of the important links in the calculation of UTC and International Atomic Time (International Atomic Time, TAI)[1-2]. At present, the punctual laboratories participating in UTC/TAI calculations mainly use GPS precise point positioning (GPS PPP), GPS all-in-view (GPS AV) and two-way satellite time and frequency transfer (TWSTFT and GPS). PPP (combined smoothing of TWSTFT and GPS PPP, TWGPPP) links, where GPS PPP links account for about 60% of the total number of time-borne links. In 2020, the International Bureau of Authority (BIPM) used the Galileo system as a backup time transmission link for UTC calculation, and the time transmission performance of the Beidou system was also tested and evaluated.

In order to ensure the accuracy and long-term stability of the GNSS time transfer link, the hardware delay of the receivers used at both ends of the link needs to be calibrated regularly. The hardware delay calibration of the receiver is to measure the hardware delay from the phase center of the antenna to the reference point of the local signal, and its value is related to the frequency of the satellite signal, mainly including the reference delay (REFDLY) between the local signal reference point and the internal reference point of the receiver, the internal delay (INTDLY) of the satellite signal inside the receiver and the antenna, and the signal group delay (cable) inside the an-

tenna cable delay[3-4]. The receiver hardware delay calibration method is divided into direct calibration and relative calibration. The U.S. Naval Research Laboratory and the French Space Agency used different types of equipment (satellite simulators, vector network analyzers) and methods to directly calibrate the delay of GPS signals inside the receiver and antenna, respectively, and the results were of good consistency, with a difference of no more than 1 ns[5]. At present, the receivers of each punctual laboratory are mainly measured by BIPM and Regional Metrology Organizations (RMO) in a relatively calibrated manner[6]. In recent years, there have been relatively few studies on the calibration of the Beidou signal delay of the receiver by scholars at home and abroad, and the calibration results have not been recognized by BIPM. In this paper, the relative calibration test of BDS PPP link was carried out in the National Time Service Center (NTSC) of the Chinese Academy of Sciences (NTSC) using GPS/BDS PPP software and an uncalibrated multimode receiver, and the total time delay of the BDS B1I&B3I signal at both ends of the receiver BDS B1I &B3I between UTC (NTSC) and UTC (JATC) (Joint Atomic Time of China) maintained by the National Time Service Center can be indirectly obtained through relative calibration delay, TOTDLY), and then get the absolute deviation between UTC (NTSC) and UTC (JATC). The accuracy of the time difference results obtained by the GPS PPP link published by BIPM was used as a reference to evaluate the accuracy of the time difference results.

2 BDS PPP link relative calibration principle

The difference between the two receivers without calibrating the Beidou signal delay can be calculated according to the principle of GNSS co-clocking comparison. If the difference between the total time delay of the Beidou signal of the receiver to be calibrated in the two punctual laboratories and the same mobile station is muted with each other, the absolute time difference of the reference time of the two places can be indirectly obtained. Taking a BDS PPP link as an example, the relative calibration process is as follows.

①As shown in Fig.1, a multi-mode receiver R_T is selected as the mobile station, and it is placed in the punctuality laboratory A and the receiver R_A to be calibrated are both connected to the local time-frequency reference UTC (A) frequency signal and synchronous observation. The PPP method was used to process the Beidou data observed by the two receivers to solve the receiver clock error. The value of the clock difference solved for the i th epoch can be written as follows:

$$T_{R_A,i} = UTC(A) - T_{ref} + D_{R_A,i} \quad (1)$$

$$T_{R_T,i} = UTC(A) - T_{ref} + D_{R_T,i} \quad (2)$$

In equations (1) and (2), $T_{R_A,i}$ and $T_{R_T,i}$ are the clock deviations calculated by the PPP method to process the Beidou observation data of the R_A and R_T receivers in the i -th epoch, respectively. $D_{R_A,i}$ and $D_{R_T,i}$ are the total BDS signal delays of the i -th epoch R_A and R_T receivers, respectively. T_{ref} is a reference clock defined by the analysis centers of IGS (International GNSS Service). Subtracting the two equations yields

the difference between the total delay of the i -th epoch R_A and the BDS signal of the R_T receiver.

$$D_{R_{A,i}} - D_{R_{T,i}} = T_{R_{A,i}} - T_{R_{T,i}} \quad (3)$$

②By moving the mobile station R_T and its antenna and connecting cables to the punctual laboratory B and the receiver R_B to be calibrated, both are connected to the frequency signal generated by the local time-frequency reference UTC (B) and observed synchronously, the total delay difference between the j th epoch R_B and the BDS signal of the R_T receiver can be obtained.

$$D_{R_{B,i}} - D_{R_{T,i}} = T_{R_{B,i}} - T_{R_{T,i}} \quad (4)$$

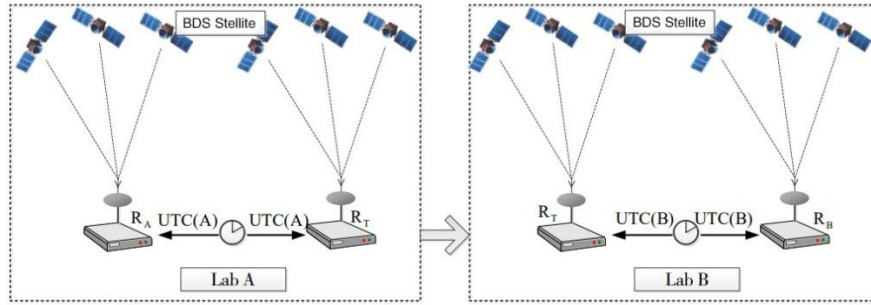


Fig. 1. BDS PPP link relative calibration principle

③Calculate the difference between the total delay of the BDS signal from the R_A and R_B receivers. Due to the short time interval between two synchronous observations, it can be assumed that the total delay of the BDS signal from the receiver used remains unchanged. However, due to the influence of random errors such as observation noise and atmospheric delay errors, there are slight differences in the clock errors of each receiver solved using PPP over time. Therefore, the difference in the total delay of the BDS signal between R_A and R_B receivers can be expressed as.

$$D_{R_A} - D_{R_B} = \sum_{n=1}^N (T_{R_{A,n}} - T_{R_{T,n}}) / N - \sum_{m=1}^M (T_{R_{B,m}} - T_{R_{T,m}}) / M \quad (5)$$

$T_{R_{A,n}}$ 、 $T_{R_{T,n}}$ represent separately the n th epoch clock error with R_A and R_T during the synchronous observation period, $T_{R_{B,m}}$ 、 $T_{R_{T,m}}$ represent separately the m th epoch clock error with R_B and R_T during the synchronous observation period, N 、 M represent the total epoch.

④Calculate the time difference between reference times in locations A and B. Due to the fact that the receivers R_A and R_B located in A and B maintain synchronous observation throughout the entire observation process, their clock deviations in the k -th epoch can be represented as.

$$T_{R_{A,k}} = UTC(A) - T_{ref} + D_{R_A} \quad (6)$$

$$T_{R_{B,k}} = UTC(B) - T_{ref} + D_{R_B} \quad (7)$$

Subtract equations (6) and (7) and substitute the result calculated by equation (5) into equation (8) to obtain the time difference between the reference times of the k -th epoch A and B, i.e.

$$(UTC(A) - UTC(B))_k = T_{R_{A,k}} - T_{R_{B,k}} - (D_{R_A} - D_{R_B}) \quad (8)$$

3 BDS PPP link relative calibration

3.1 GPS/BDS PPP Software evaluation

The use of GPS PPP method for high-precision time transfer results in lower link noise compared to co view and full view methods. Therefore, in order to improve the accuracy of Beidou link calibration, GPS/BDS PPP software was developed and its reliability was verified. Multiple mode and multi frequency receivers PT13 and TP02 with local reference signals were selected from the German Institute of Technical Physics (PTB) and the Czech Institute of Optoelectronics (TP), respectively. The sampling interval for their observation data is 30 seconds, and the specific configuration information of the receivers is shown in Table 1. Using GPS/BDS PPP software, the dual frequency carrier and pseudorange observations of GPS and BDS were processed separately to obtain the clock deviation of the reference time between the two locations, as shown in Fig.2. Table 2 shows the models and strategies adopted by PPP software in the data processing process. Evaluate the time comparison results using the GPS PPP results published by BIPM as a reference.

Table 1. Receiver configuration information

laboratory	Receiver number	Receiver type	Antenna type
PTB	PT13	POLARX5TR	LEIAR25.R4
TP	TP02	GTR55	NOV850

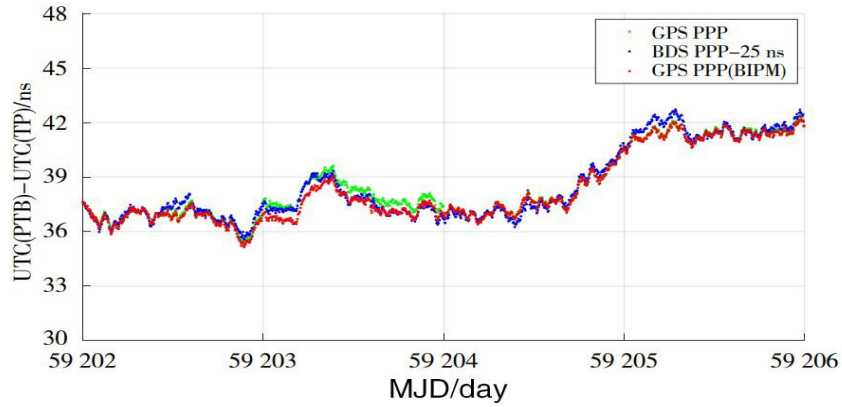


Fig. 2. Time comparison results between PTB and TP

Table 2. GPS/BDS observation data processing models and strategies

Content	models and strategies
Observation model	Traditional models
Observation signal	GPS L1、L2;BDS B1I、 B3I
Satellite orbit、 error	WUM Precision products
Ionosphere	dual frequency ionospheric combination

Troposphere
parameter estimation

Model + Random walk noise
Weighted Least Squares

From Fig.2, it can be seen that the time comparison results of GPS and BDS PPP links calculated using development software are consistent with the GPS PPP results published by BIPM. Due to the uncalibration of the BDS link, there is a significant system bias in the time comparison results, but according to Section 2, this does not affect the relative calibration results of the BDS PPP link. Based on the comparison results of BIPM GPS PPP links, the root mean square (RMS) values of GPS and BDS PPP links relative to BIPM GPS PPP links during the entire observation period were 0.29 ns and 25.17 ns, respectively. It can be considered that the processing results of GPS/BDS PPP software are reliable.

3.2 BDS PPP analysis of link relative calibration results

The 5 multimode receivers used in the experiment were all from the National Time Service Center, and the signal delay of each satellite system was not calibrated. The specific configuration information is shown in Table 3. The time frequency reference signal is UTC(NTSC) of NT02、NT03 and NT05 , the time frequency reference signal is UTC(JATC) of JA02. To verify the feasibility of the BDS link relative calibration method mentioned above, the method was evaluated with a common clock difference (CCD) result of 0 as a reference. And use the developed PPP software to process the BDS observation data of each receiver and solve the clock error. Fig. 3 shows common clock comparison results of the BDS PPP link between any two receivers.

Table 3. Receiver configuration information

Receiver number	Receiver type	Antenna type
NT02	GTR55	JAVADTRE_03
NT02	GTR55	NOV850
NT05	POLARX5TR	SEPCHOKE_B36E
NT06	POLARX5TR	SEPCHOKE_B36E
JA02	GTR55	NOV850

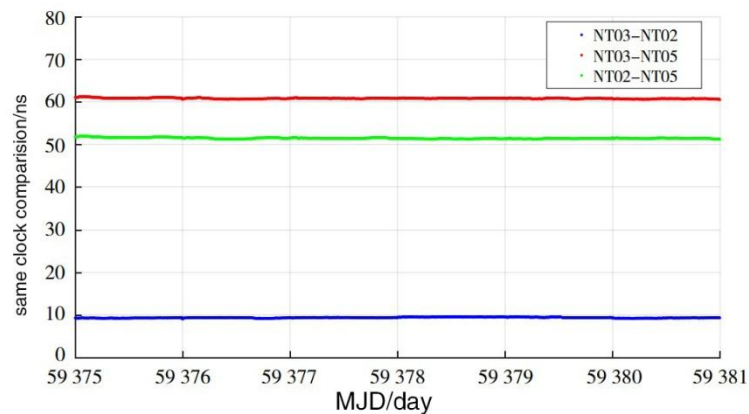


Fig. 3. BDS PPP link Time comparison results(uncalibrated)

Due to the uncalibrated Beidou signal delay of each receiver, the BDS PPP link clock comparison results are not around 0. If NT05 is set as a mobile station, the total delay difference between NT03 and NT02 receiver BDS signals can be calculated as 9.36 ns according to formulas (1) to (5). Fig. 4 shows the clock comparison results of the BDS PPP link after deducting the difference in the total delay of the BDS signals from the NT03 and NT02 receivers. Within the range of ± 0.2 ns, the RMS relative to the reference value of 0 is approximately 0.1 ns.

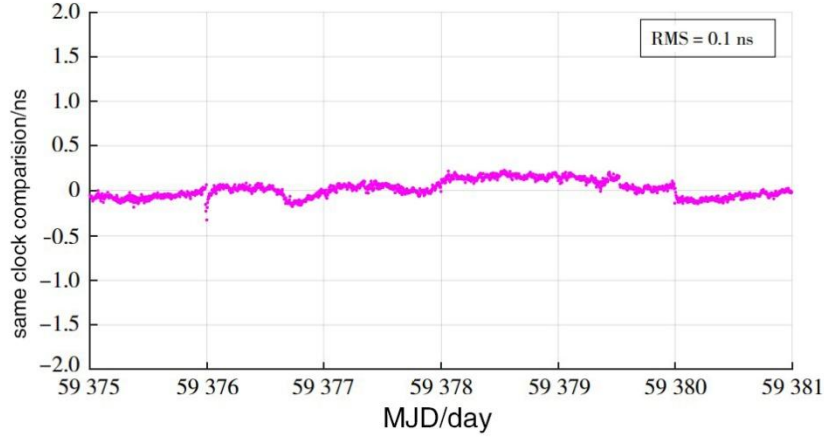


Fig. 4. BDS PPP link Time comparison results between NT02 and NT03(calibrated)

To further validate the effectiveness of the calibration method, the GPS PPP results published by BIPM were used as a reference to evaluate the time comparison results of the BDS PPP link relative calibration between UTC (NTSC) and UTC (JATC). Set NT06 as a mobile station, first connect it to the UTC (NTSC) time-frequency reference signal for synchronous observation with NT03, and then connect NT06 to the UTC (JATC) time-frequency reference signal for synchronous observation with JA02. Throughout the entire observation process, the antenna, cable, and time-frequency reference signal line of NT06 remain unchanged to ensure stable delay in all parts.

Fig. 5 and 6 respectively show the comparison results of BDS PPP link clock sharing between NT06 and NT03, as well as between NT06 and JA02. According to formulas (1) to (5), the total delay difference of BDS signal between NT03 and JA02 can be calculated as 83.53 ns. Fig. 7 shows the time comparison results of the BDS PPP link between UTC (NTSC) and UTC (JATC) after relative calibration. The peak residual relative to the GPS PPP results published by BIPM is within ± 0.2 ns, as shown in Fig. 8.

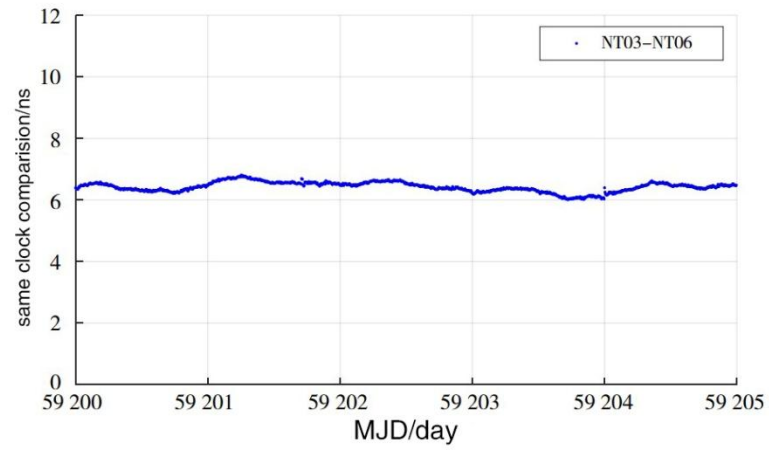


Fig. 5. BDS PPP link Time comparison results between NT03 and NT06(uncalibrated)

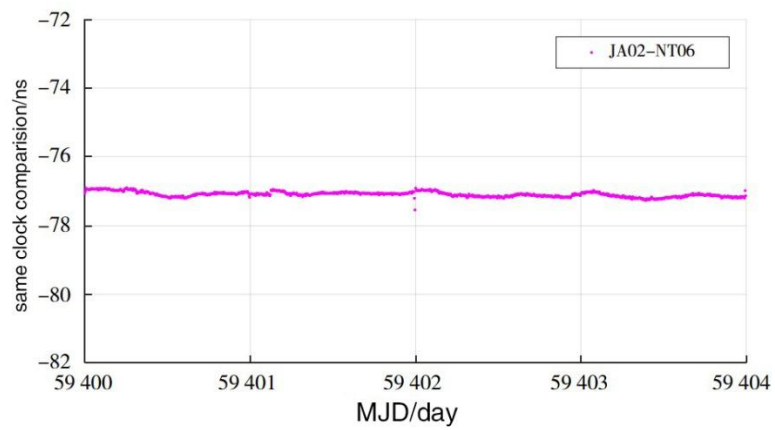


Fig. 6. BDS PPP link Time comparison results between JA02 and NT06(uncalibrated)

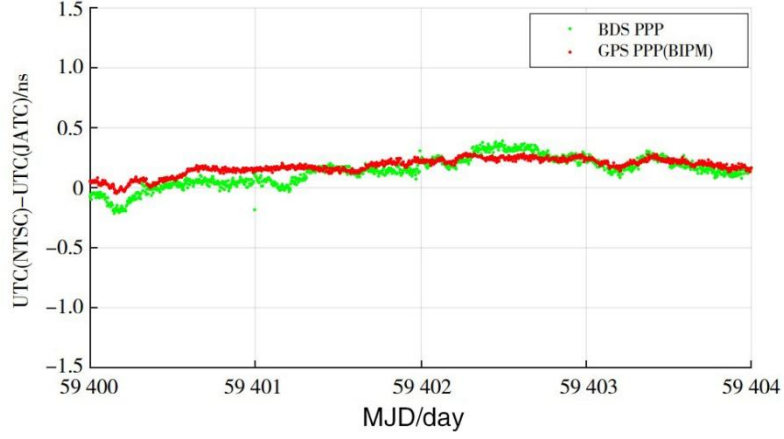


Fig. 7. BDS PPP link Time comparison results between UTC(NTSC) and UTC(JATC)(uncalibrated)

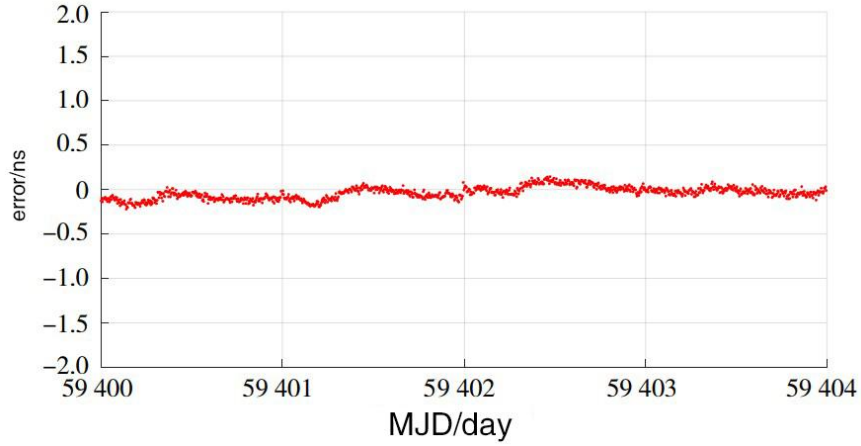


Fig. 8. BDS PPP link Time comparison error between UTC(NTSC) and UTC(JATC)

4 Conclusion

This article conducted BDS PPP link relative calibration experiments using self-developed GPS/BDS PPP software and multiple uncalibrated GNSS multimode receivers. The time comparison results of BDS PPP link relative calibration between UTC (NTSC) and UTC (JATC) are basically consistent with the GPS PPP link comparison results published by BIPM, with a peak deviation of no more than 0.2 ns, meeting the accuracy requirements of GNSS time transmission, proving the feasibility of this calibration method.

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